

ABC Classification and Inventory Costs

Selected solutions from Lecture 01

Contents

1. Part 4 · How do we begin? — ABC analysis — lecture example (10 part numbers)
2. Part 4 · How do we begin? — ABC classification of 200 SKUs
3. Part 6 · Show me the money — Carrying cost: from the annual rate r to the unit holding cost $h = r \cdot v$
4. Part 6 · Show me the money — The holding vs ordering cost trade-off (preview of lot sizing)

1. ABC analysis — lecture example (10 part numbers)

Step 1 — annual dollar usage

Annual usage analysis (slide 32).

part number	annual unit usage	unit cost	annual dollar usage
1	1,100	\$2	\$2,200
2	600	\$40	\$24,000
3	100	\$4	\$400
4	1,300	\$1	\$1,300
5	100	\$60	\$6,000
6	10	\$25	\$250
7	100	\$2	\$200
8	1,500	\$2	\$3,000
9	200	\$2	\$400
10	500	\$1	\$500

\$38,250

Total annual \$ usage

10

Number of parts

Step 2 — rank, cumulate and classify

Ranking by annual \$ usage (slide 34) with classes from both cut-off rules.

part number	annual dollar usage	cum value	cum pct value	cum pct items	class	class by value 80 95
2	\$24,000	\$24,000	62.7%	10%	A	A
5	\$6,000	\$30,000	78.4%	20%	A	A
8	\$3,000	\$33,000	86.3%	30%	B	B
1	\$2,200	\$35,200	92.0%	40%	B	B
4	\$1,300	\$36,500	95.4%	50%	B	C
10	\$500	\$37,000	96.7%	60%	C	C
9	\$400	\$37,400	97.8%	70%	C	C
3	\$400	\$37,800	98.8%	80%	C	C
6	\$250	\$38,050	99.5%	90%	C	C
7	\$200	\$38,250	100.0%	100%	C	C

With the **items rule (20 % / 30 % / 50 %)** parts 2 and 5 are A, parts 8, 1, 4 are B and the rest C — A holds **78 %** of the value with 20 % of the items, exactly the Pareto pattern on slide 30. With **strict value cut-offs (80 % / 95 %)** part 4 (cumulative 95.4 %) falls just into C: items at a boundary are a **judgement call** — managers often promote them for other reasons (criticality, scarcity).

Pareto chart — lecture example

Part 2 alone is 63 % of annual \$ usage

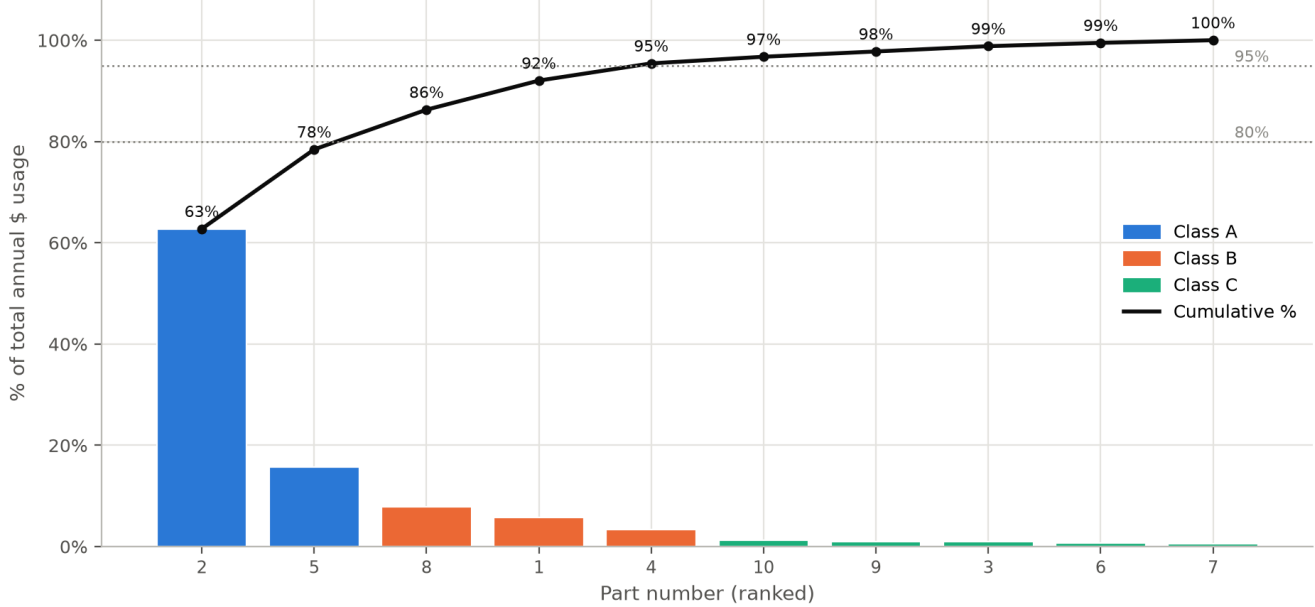


Figure 1.1 — Bars = % of annual \$ usage per part (coloured by class); line = cumulative %.

ABC curve

Cumulative % of items vs cumulative % of value

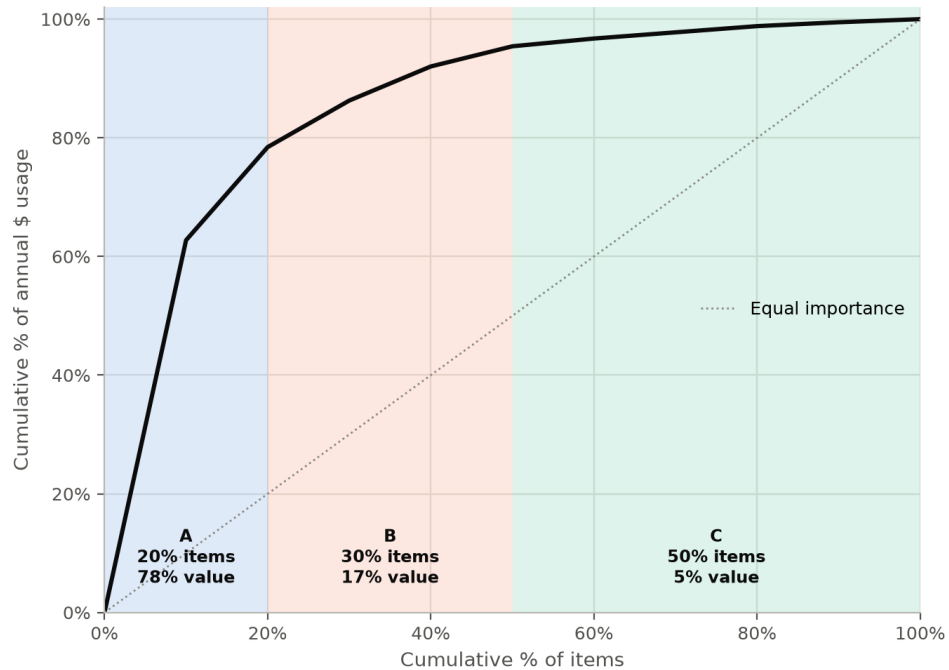


Figure 1.2 — ABC (Lorenz) curve with class bands.

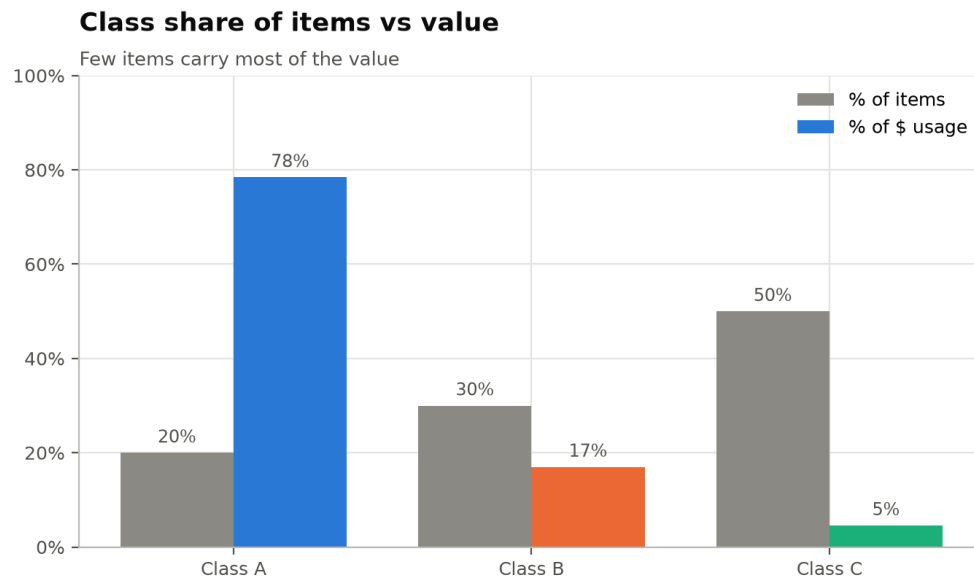


Figure 1.3 — Share of items and value by class.

Class summary (items rule).

class	items	value	pct items	pct value
A	2	\$30,000	20%	78.4%
B	3	\$6,500	30%	17.0%
C	5	\$1,750	50%	4.6%

2. ABC classification of 200 SKUs

Classification rule: cumulative % of value — A ≤ 80%, B ≤ 95%, C = rest.

\$1,832,644 Total annual \$ usage	45 items (22%) → 79.8% of value Class A	64 items (32%) → 15.1% of value Class B
91 items (46%) → 5.1% of value Class C	SKU-0172 = 19.4% of value Top SKU	0.75 Concentration (Gini)

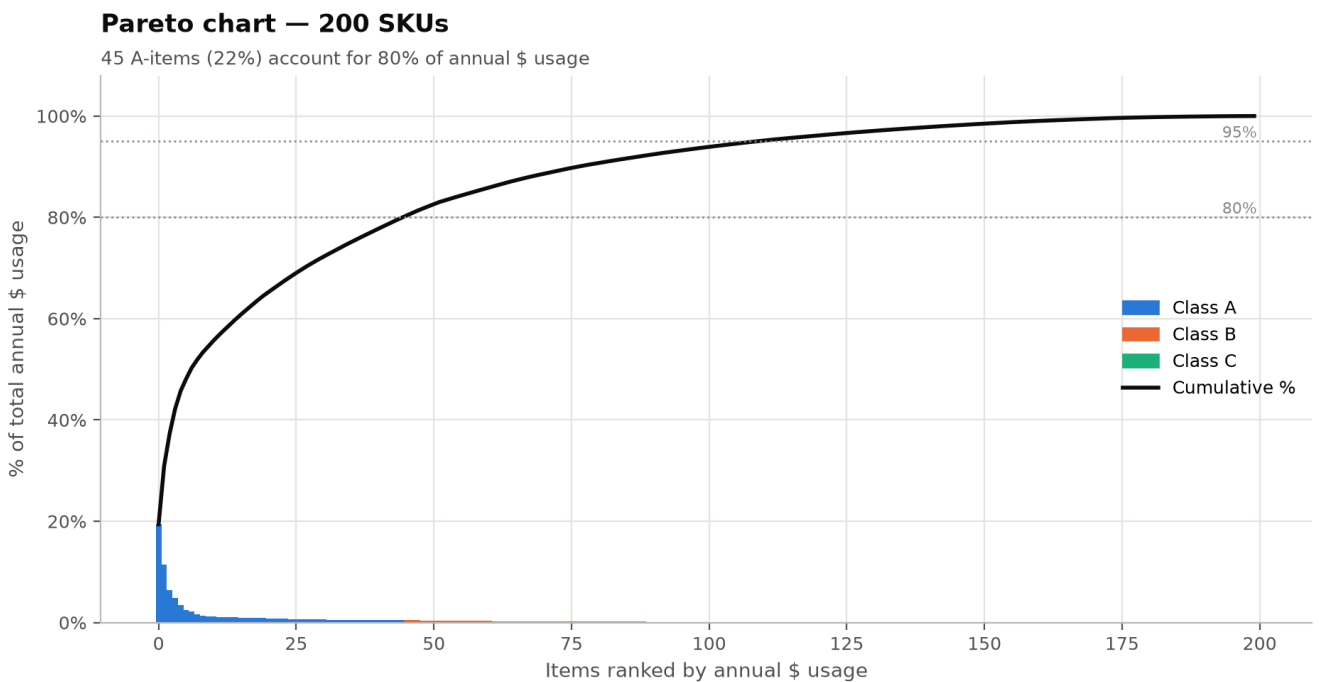


Figure 2.1 — Pareto chart of all 200 SKUs coloured by class.

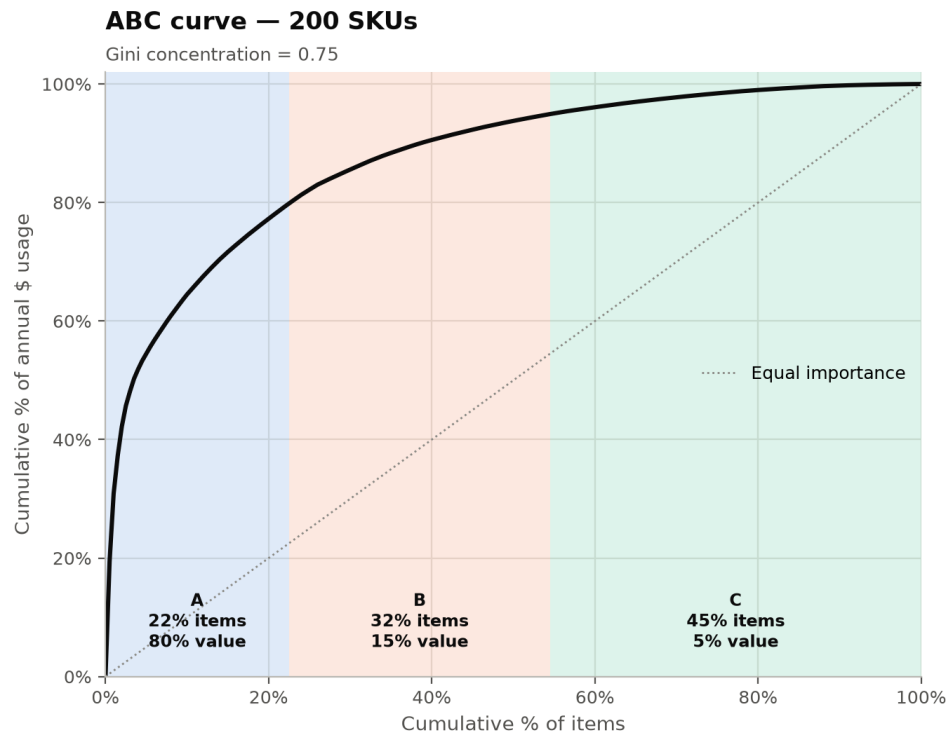
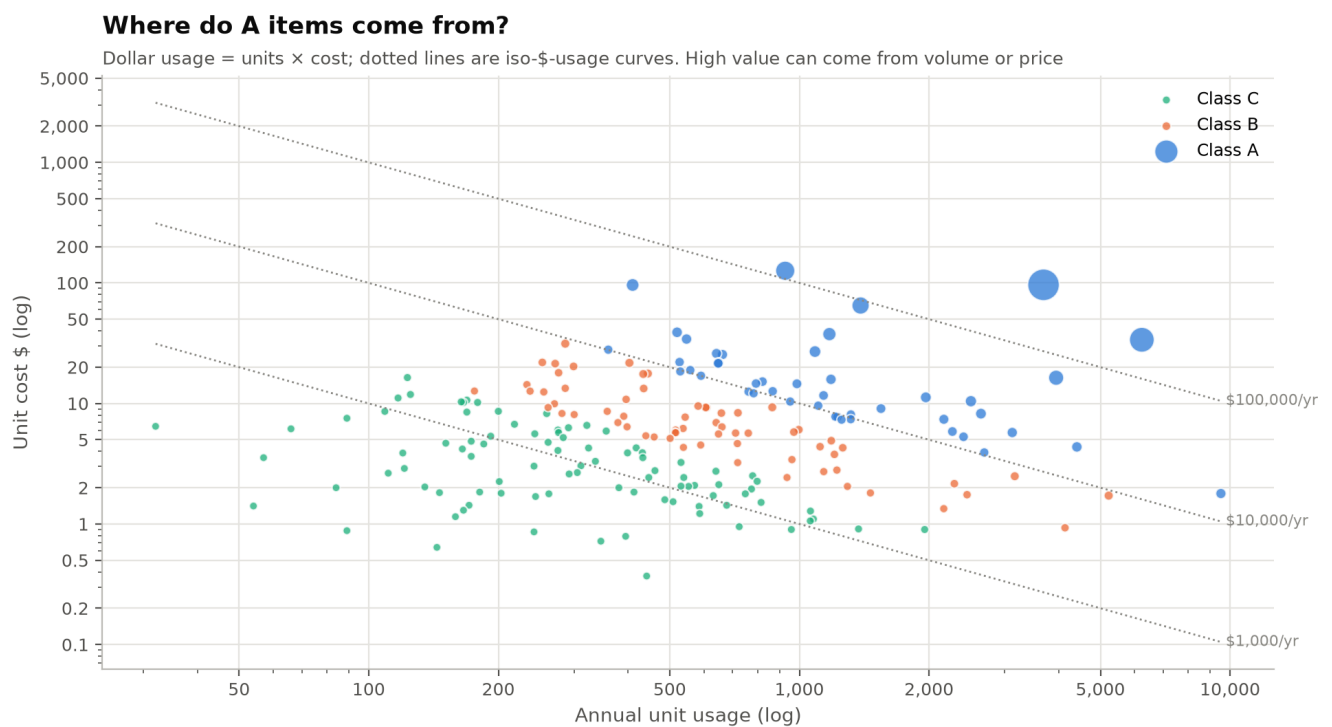


Figure 2.2 — ABC (Lorenz) curve.

Figure 2.3 — Each bubble is a SKU; bubble area $\propto$ annual \$ usage.

ABC mix by item category

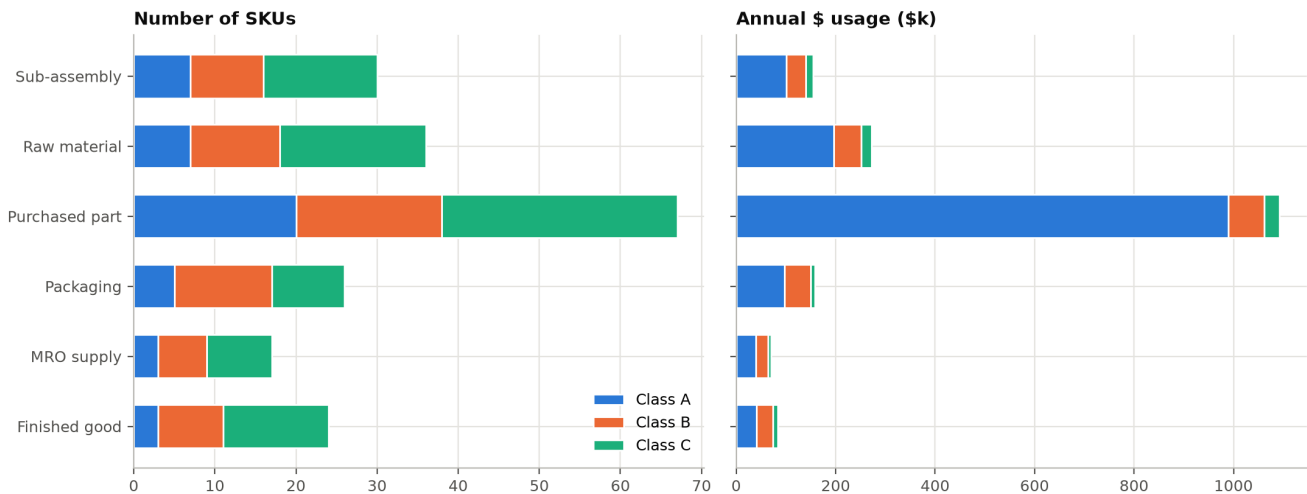


Figure 2.4 — Count and value of SKUs by category and class.

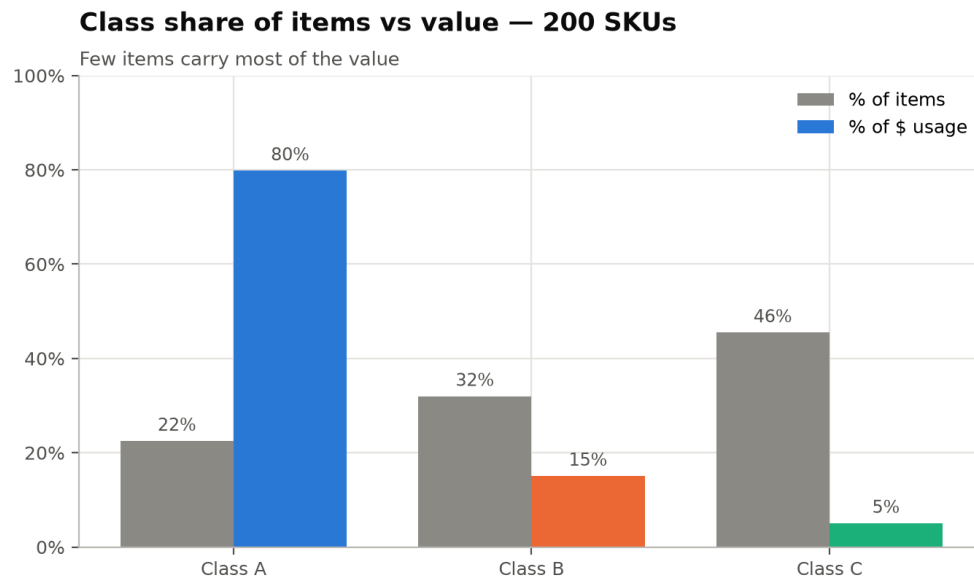


Figure 2.5 — Class summary.

Class summary.

class	items	value	pct items	pct value
A	45	\$1,463,320	22.5%	79.8%
B	64	\$276,463	32.0%	15.1%
C	91	\$92,861	45.5%	5.1%

Ranked SKU list (first 40 rows; full list in reports/tables).

rank	sku	category	annual unit usage	unit cost	annual dollar usage	pct of value	cum pct value	cum pct items	class
1	SKU-0172	Purchased part	3,689	\$96.42	\$355,693	19.41%	19.4%	0.5%	A
2	SKU-0052	Purchased part	6,240	\$33.75	\$210,600	11.49%	30.9%	1.0%	A
3	SKU-0123	Purchased part	927	\$126.14	\$116,932	6.38%	37.3%	1.5%	A
4	SKU-0176	Raw material	1,387	\$64.94	\$90,072	4.91%	42.2%	2.0%	A
5	SKU-0046	Purchased part	3,946	\$16.34	\$64,478	3.52%	45.7%	2.5%	A
6	SKU-0024	Raw material	1,174	\$37.61	\$44,154	2.41%	48.1%	3.0%	A

rank	sku	category	annual unit usage	unit cost	annual dollar usage	pct of value	cum pct value	cum pct items	class
7	SKU-0041	Packaging	410	\$96.15	\$39,422	2.15%	50.3%	3.5%	A
8	SKU-0019	Purchased part	1,087	\$26.94	\$29,284	1.60%	51.9%	4.0%	A
9	SKU-0043	Purchased part	2,502	\$10.44	\$26,121	1.43%	53.3%	4.5%	A
10	SKU-0192	Sub-assembly	1,966	\$11.24	\$22,098	1.21%	54.5%	5.0%	A
11	SKU-0117	Raw material	2,640	\$8.23	\$21,727	1.19%	55.7%	5.5%	A
12	SKU-0112	Purchased part	520	\$38.94	\$20,249	1.10%	56.8%	6.0%	A
13	SKU-0069	Packaging	4,409	\$4.36	\$19,223	1.05%	57.8%	6.5%	A
14	SKU-0130	Sub-assembly	1,184	\$15.85	\$18,766	1.02%	58.9%	7.0%	A
15	SKU-0072	Purchased part	547	\$34.22	\$18,718	1.02%	59.9%	7.5%	A
16	SKU-0185	Purchased part	3,121	\$5.74	\$17,915	0.98%	60.9%	8.0%	A
17	SKU-0109	Sub-assembly	9,523	\$1.79	\$17,046	0.93%	61.8%	8.5%	A
18	SKU-0134	Finished good	663	\$25.46	\$16,880	0.92%	62.7%	9.0%	A
19	SKU-0016	MRO supply	642	\$25.97	\$16,673	0.91%	63.6%	9.5%	A
20	SKU-0045	Purchased part	2,165	\$7.39	\$15,999	0.87%	64.5%	10.0%	A
21	SKU-0082	Packaging	987	\$14.56	\$14,371	0.78%	65.3%	10.5%	A
22	SKU-0111	Packaging	649	\$21.75	\$14,116	0.77%	66.1%	11.0%	A
23	SKU-0091	Finished good	1,547	\$9.07	\$14,031	0.77%	66.8%	11.5%	A
24	SKU-0005	Purchased part	648	\$21.52	\$13,945	0.76%	67.6%	12.0%	A
25	SKU-0158	Purchased part	1,138	\$11.65	\$13,258	0.72%	68.3%	12.5%	A
26	SKU-0083	Purchased part	2,266	\$5.85	\$13,256	0.72%	69.0%	13.0%	A
27	SKU-0141	Sub-assembly	2,405	\$5.29	\$12,722	0.69%	69.7%	13.5%	A
28	SKU-0179	MRO supply	821	\$15.18	\$12,463	0.68%	70.4%	14.0%	A
29	SKU-0099	Purchased part	527	\$22.04	\$11,615	0.63%	71.0%	14.5%	A
30	SKU-0085	Raw material	793	\$14.63	\$11,602	0.63%	71.7%	15.0%	A
31	SKU-0200	Purchased part	867	\$12.63	\$10,950	0.60%	72.3%	15.5%	A
32	SKU-0053	Purchased part	1,317	\$8.10	\$10,668	0.58%	72.8%	16.0%	A
33	SKU-0110	Purchased part	1,106	\$9.58	\$10,595	0.58%	73.4%	16.5%	A
34	SKU-0002	Sub-assembly	558	\$18.90	\$10,546	0.58%	74.0%	17.0%	A
35	SKU-0164	MRO supply	2,687	\$3.92	\$10,533	0.57%	74.6%	17.5%	A
36	SKU-0156	Raw material	360	\$27.93	\$10,055	0.55%	75.1%	18.0%	A
37	SKU-0066	Finished good	591	\$16.97	\$10,029	0.55%	75.7%	18.5%	A
38	SKU-0054	Purchased part	952	\$10.37	\$9,872	0.54%	76.2%	19.0%	A
39	SKU-0026	Purchased part	529	\$18.45	\$9,760	0.53%	76.7%	19.5%	A
40	SKU-0097	Sub-assembly	1,316	\$7.40	\$9,738	0.53%	77.3%	20.0%	A

... 160 more rows not shown — full table exported to reports/tables/.

3. Carrying cost: from the annual rate r to the unit holding cost $h = r \cdot v$

$$r = 28\% + 2\% + 6\% + 1\% = 37\% \quad h = 0.37 \times 180 = 66.60 \quad \text{Total} = 5 \times 300 \times 66.60 = 99,900.00$$

37 % Annual carrying rate r	37 Piasters per EGP per year	\$66.60 /unit/yr h for the \$180 item
\$99,900 300 units $\times$ 5 years	76% Capital cost share of r	

Carrying cost — $r = 37\%$ per year (slide 45)

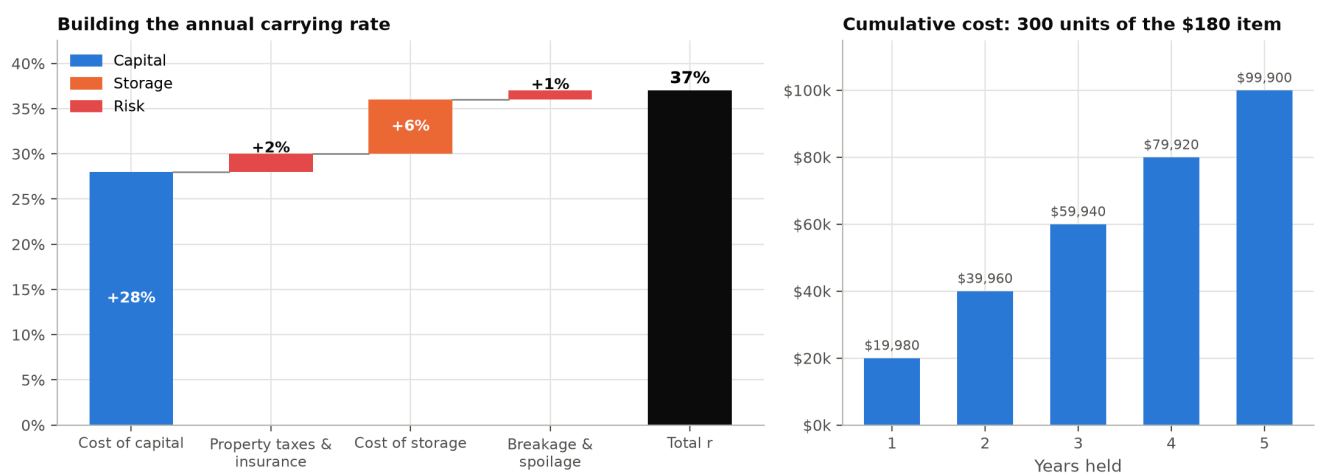


Figure 3.1 — Left: components of r (waterfall). Right: cumulative holding cost of the lecture example.

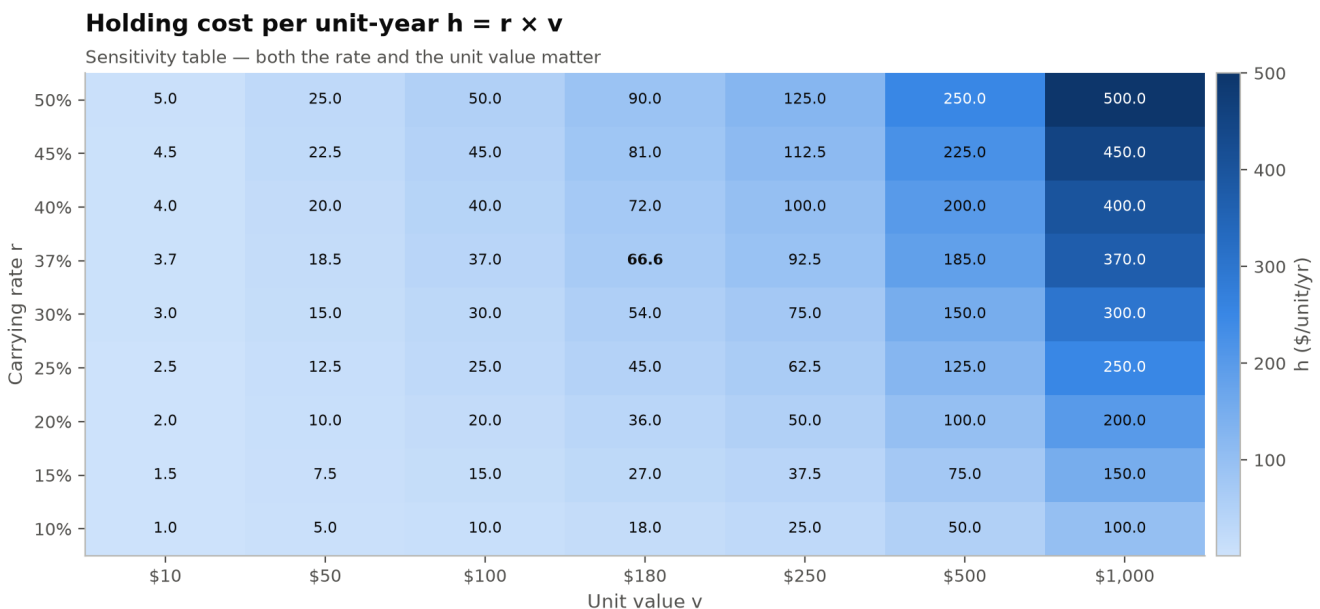


Figure 3.2 — h for combinations of r and v .

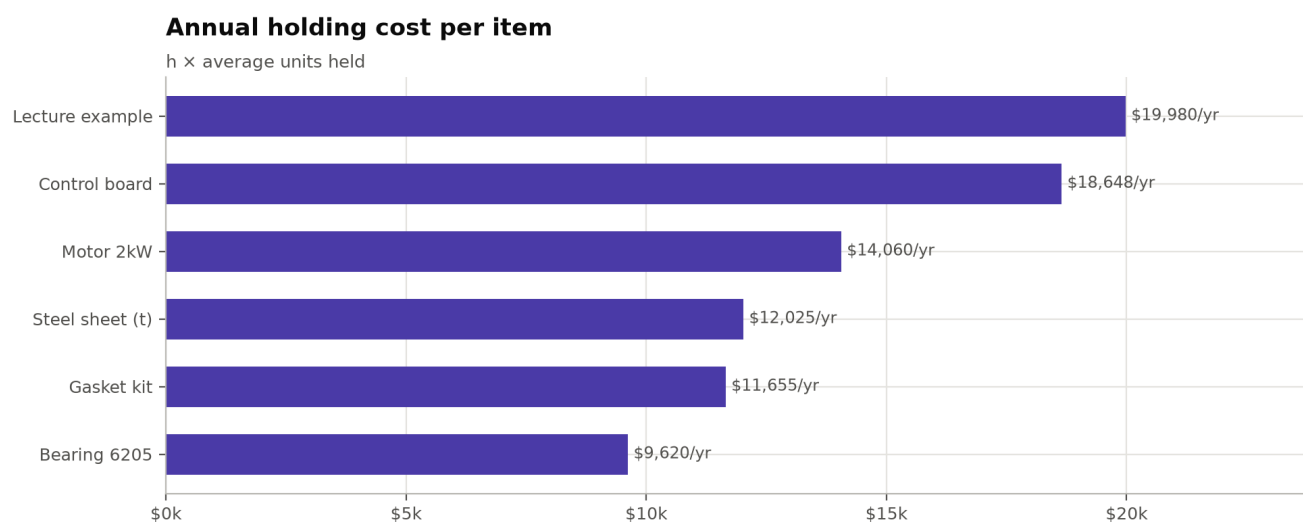


Figure 3.3 — Annual holding cost by item at $r = 37\%$.

Carrying-rate components.

component	group	annual rate pct
Cost of capital	Capital cost	28
Property taxes & insurance	Risk cost	2
Cost of storage	Storage cost	6
Breakage & spoilage	Risk cost	1

Holding cost by item.

item	unit value	avg units held	years held	h per unit year	holding cost per year	total holding cost
Lecture example	\$180.00	300	5	\$66.60	\$19,980	\$99,900
Motor 2kW	\$950.00	40	1	\$351.50	\$14,060	\$14,060
Control board	\$420.00	120	1	\$155.40	\$18,648	\$18,648
Steel sheet (t)	\$1,300.00	25	1	\$481.00	\$12,025	\$12,025
Gasket kit	\$35.00	900	1	\$12.95	\$11,655	\$11,655
Bearing 6205	\$6.50	4,000	1	\$2.40	\$9,620	\$9,620

4. The holding vs ordering cost trade-off (preview of lot sizing)

Item **Inverter PCB** — $D = 4,800/\text{yr}$, $v = 410$, $A = 2,630$ per order, $r = 37\% \rightarrow h = 151.70$ per unit-year.

$$\text{TRC}(Q) = A \cdot D/Q + h \cdot Q/2$$

408 units

Cost-minimising Q (numerical)

11.8

Orders per year

61,888/yr

Minimum total relevant cost

30,941 vs 30,947

Ordering = holding at optimum?

262–635

Cost within +10 % for Q in

Answering 'how large should the order be?'

Minimum total cost where ordering and holding costs balance — and the curve is flat near the optimum

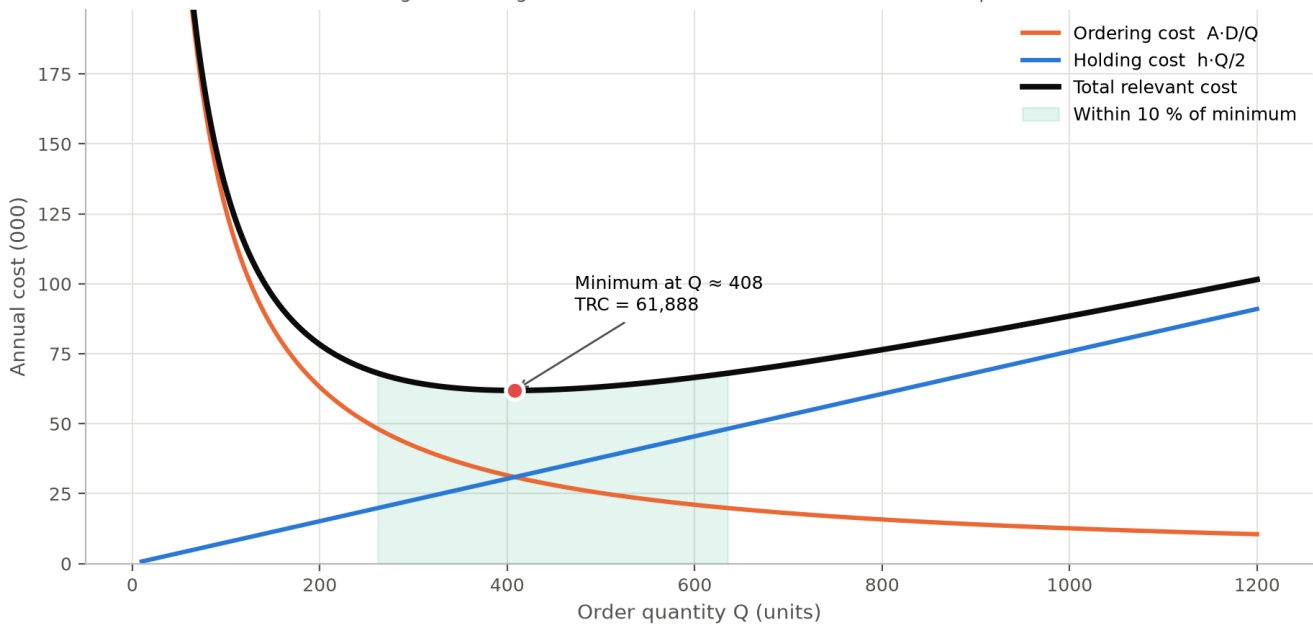


Figure 4.1 — Annual ordering, holding and total relevant cost vs order quantity.

Cost of alternative order quantities.

Q	orders per year	avg cycle stock	ordering cost	holding cost	total cost	pct above min
50	96.0	25	252,480	3,792	256,272	+314.1%
100	48.0	50	126,240	7,585	133,825	+116.2%
200	24.0	100	63,120	15,170	78,290	+26.5%
408	11.8	204	30,941	30,947	61,888	+0.0%
400	12.0	200	31,560	30,340	61,900	+0.0%
800	6.0	400	15,780	60,680	76,460	+23.5%